USRP X310 Data Recording Report

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Abstract

This report describes the data recording process using the USRP X310 device. This reciver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)[1]. The present report is updated as new recordings are made and new information is available.

1 Overview of the recordings

The recording is computed automatically by a Linux service which is driven by the following guidelines:

- Everyday, the program will start within the time rage [4PM, 6PM]
- It will sense whether the USRP is detectable via `uhd_find_devices --args=''addr=192.168.10.2''`.
- If not found, it will exit silently and will sense again in intervals of 15 min.
- Once found, a SigMF file will be created from the recording, with the interleaved I/Q samples + metadata, like number of channels (dual channel recording), frequency (L1 and L5), sampling rate, gain, sample type (signed 16-bit int), etc. Each channel has its own configuration (see the table below), so the metadata of each channel is reported separately.
  - Channel 0 (RX0): GPS L1 (1575.42 MHz), 5 MS/s, 25 dB of gain;
  - Channel 1 (RX1): GPS L5 (1176.45 MHz), 20 MS/s, 25 dB of gain.

  Records made before 17/10/2026 used 2 MS/s in both channels.
- Each record lasts 5 minutes (300 s)
- It will record a second time within the same range and 30 min after the first record, and then it will not record anymore for that day.

[1] https://github.com/ita-gnss-lab

- After 15 days of records from the last report. Rubem will send an automatic message to header of the Laboratory, Prof. Dr. Felix Antreich, via WhatsApp with the summarized information from all recorded metadata;

October

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Table 1: Recording calendar. Green: 2 recordings made; blue: 1 recording made. No color: no recording made.

2 Recording reports

2.1 Records on 08/10/2026

2.1.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-08 22:26:37 UTC	
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Frequency	1575.420 MHz	1176.450 MHz
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 2: Metadata for recording `x310_20261008T222630Z`, captured starting 2026-10-08T22:26:37.215640Z.

2.2 Records on 13/10/2026

2.2.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-13 21:57:56 UTC	
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Frequency	1575.420 MHz	1176.450 MHz
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 3: Metadata for recording `x310_20261013T215749Z`, captured starting 2026-10-13T21:57:56.746726Z.

2.2.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-13 22:08:41 UTC	
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Frequency	1575.420 MHz	1176.450 MHz
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 4: Metadata for recording `x310_20261013T220835Z`, captured starting 2026-10-13T22:08:41.589855Z.